Commercial Auto Loss Reserving in SQL (DuckDB)

Status: in progress. Data preparation, company screening, triangle construction, and development factors are complete. Reserve estimation and back-testing are under way.

Business question

As of the end of 2007, how much should a mid-sized commercial auto insurer hold in loss reserves for accident years 1998–2007, how uncertain is that estimate, and how well would it have predicted actual development afterwards?

Data

- CAS *Loss Reserving Data Pulled From NAIC Schedule P* (commercial auto file), downloaded from the Casualty Actuarial Society website. The raw file is not included in this repository.
- 157 companies; each complete company has 100 rows (10 accident years x 10 development lags, calendar years 1998–2016).
- Valuation date: end of 2007.
  - **Upper triangle** (calendar year <= 2007): the information known at the valuation date, used for modelling.
  - **Lower triangle** (calendar years 2008–2016): reserved for back-testing only.

Data dictionary

Column	Meaning
GRCODE / GRNAME	Company code and name
AccidentYear	Year the loss occurred
DevelopmentYear	Calendar year of the observation (= AccidentYear + DevelopmentLag - 1)
DevelopmentLag	Development age in years (1–10)
IncurredLosses	Incurred loss (paid + case reserves + bulk reserves)
CumPaidLoss	Cumulative paid loss
BulkLoss	Bulk reserve (IBNR and other reserves not tied to individual cases)
EarnedPremDIR / EarnedPremCeded / EarnedPremNet	Direct, ceded and net earned premium
Single	Single-entity indicator (1 = single company, 0 = group)
PostedReserves2007	Reserve posted by the company at the end of 2007

Amounts are in thousands of dollars (to be confirmed against the CAS documentation).

Company screening

Step	Companies remaining
All companies in file	157
Complete 10 x 10 data	137
No negative cumulative paid, positive net premium	95
Cumulative paid never decreases within an accident year	15

Companies with decreasing cumulative paid were excluded because of likely salvage/subrogation recoveries or data adjustments, which create age-to-age factors below 1.

I selected **GRCODE 26433 (Harco Natl Ins Co)**. Its Accident Year 1998 paid loss ratio development is smooth and flattens by lag 6, and its premium volume is mid-sized. Its net premium grew about five-fold between AY 1998 and AY 2004, and its early payment pattern appears to slow after AY 2001; I treat both as diagnostic items.

Reasonableness check

Triangle-implied reserve on the latest diagonal (incurred minus paid) is 85,508, about 6.5% below the posted reserve of 91,061. The difference may reflect a different basis or accident years before 1998 included in the posted figure.

Diagnostics

Age-to-age factors show an upward shift at early lags for recent accident years: the 1→2 factor averages about 2.0 for AY 1998–2000 versus 2.2–2.4 for AY 2003–2006, and the 2→3 factor rises from 1.27–1.33 (AY 1998–2000) to about 1.5 (AY 2002–2005). The 3→4 factor is also higher for AY 2003–2004 than for AY 1998–2002, but this rests on only two observations. From lag 4 onward there is no clear trend: the 4→5 factor fluctuates between 1.07 and 1.17, later factors are close to 1.00–1.05, and the 9→10 factor has a single observation.

This is consistent with slower early payments in recent years, although premium also grew roughly five-fold over the period, so changes in business mix or claims handling cannot be ruled out. An all-year average may therefore understate development for recent accident years; alternative averaging periods will be tested.

Project structure

```
reserving-sql-project/
├── README.md
├── .gitignore
├── sql/
│   ├── 00_load.sql              load data, sanity checks
│   ├── 01_select_company.sql    company screening
│   ├── 02_triangles.sql         upper-triangle views, posted-reserve check
│   ├── 03_factors.sql           age-to-age and volume-weighted factors
│   └── 04_cdf.sql               cumulative development factors, tail factor
├── exhibits/                    charts
└── r/                           (planned) Mack model and bootstrap
```

How to reproduce

1. Download the commercial auto file from the CAS website and place it in `data/`.
2. Install DuckDB (or DBeaver with its DuckDB driver).
3. Edit the file path in `sql/00_load.sql`, then run the scripts in numerical order.

Progress

- ☑ Data loading and sanity checks
- ☑ Company screening
- ☑ Upper-triangle construction
- ☑ Age-to-age factors and diagnostics
- ☑ Cumulative development factors (tail factor = 1.00)
- ☐ Paid chain-ladder reserve and comparison with posted reserve
- ☐ Bornhuetter-Ferguson and Cape Cod methods
- ☐ Selected reserve and sensitivity analysis (tail factor, averaging period)
- ☐ Back-test against the lower triangle
- ☐ Mack model for reserve variability (R)

Limitations

- Single company; conclusions do not generalise to other insurers.
- During company screening I viewed some lower-triangle values (lag 5 and lag 10 loss ratios), which may introduce mild look-ahead bias. The selection rationale above relies on upper-triangle information only.
- The tail factor is assumed to be 1.00 based on the flattening of the oldest accident year; sensitivity to this assumption will be tested.
- Lag 10 paid loss is used as an approximation of the ultimate in the back-test; some further development may occur after lag 10.